

Today: Equilibria and Stability Review

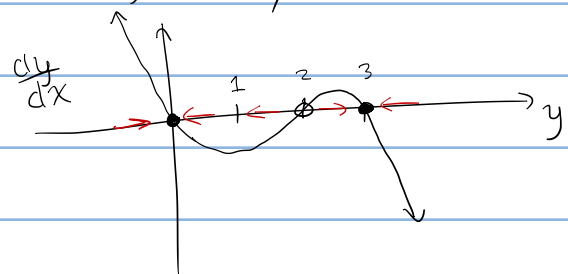
Def. If $\hat{N}$ is an equilibrium point of $\frac{dN}{dt} = g(N)$, then $g'(\hat{N})$ is called the eigenvalue of $\hat{N}$.

Ex: $\frac{dy}{dx} = y(2-y)(y-3)$

(a) Find the equilibrium points

$$\begin{aligned} y(2-y)(y-3) &= 0 \\ \swarrow \quad \quad \quad \searrow \\ y=0 \quad 2-y=0 \quad y-3=0 \\ \hat{y} &= 0, 2, 3 \end{aligned}$$

(b) Graph $\frac{dy}{dx}$ and determine stability



$$\begin{aligned} y(2-y)(y-3) \\ -y^3 + \dots \end{aligned}$$

(c) Compute eigenvalues to determine stability

$$g(y) = y(-y^2 + 3y + 2y - 6) \quad g'(0) = -6 < 0 \Rightarrow \text{stable}$$

$$= -y^3 + 5y^2 - 6y$$

$$g'(2) = -3 \cdot 4 + 10 \cdot 2 - 6 = -12 + 20 - 6 = 2 > 0 \text{ unstable}$$

$$g'(y) = -3y^2 + 10y - 6$$

$$g'(3) = -3 \cdot 9 + 10 \cdot 3 - 6 = -27 + 30 - 6 = -3 < 0 \text{ stable}$$

Review

6.3 Applications of Integration

6.3.1 Area between curves

$$\int [f(x) - g(x)] dx$$

6.3.2 Cumulative Change

$$\frac{dN}{dt} = f(t), \quad N(0) = N_0$$

$$\text{Then } N(t) = \int_0^t f(u) du + N_0$$

6.3.3 Average Values

$$\bar{f} = \frac{1}{b-a} \int_a^b f(x) dx$$

7.6 Taylor Polynomials

$$f(x) \approx \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k$$

Chapter 8 Differential Equations

8.1 Solving Differential Equations

separable $\frac{dy}{dx} = f(x)g(y) \Rightarrow \int \frac{dy}{g(y)} = \int f(x) dx$

pure time $\frac{dx}{dt} = f(t) \Rightarrow \int dx = \int f(t) dt$

autonomous $\frac{dy}{dx} = g(y) \Rightarrow \int \frac{dy}{g(y)} = \int dx$

Notable DEs are logistic equation

$$\frac{dN}{dt} = rN\left(1 - \frac{N}{K}\right)$$

r - growth rate
 K - carrying capacity

- First-order linear DEs

$$\frac{dy}{dx} + p(x)y = q(x)$$

Multiply by integrating factor $I(x) = e^{\int p(x)dx}$

$$\frac{d}{dx}(I(x)y) = I(x)q(x)$$

$$y(x) = \frac{1}{I(x)} \int I(x)q(x)dx$$

8.2 Equilibria and Stability

Autonomous DEs

$$\frac{dN}{dt} = g(N)$$

$\hat{N}$ is equilibrium point of the system if $g(\hat{N}) = 0$.

The eigenvalue associated with an equilibrium point $\hat{N}$ is $g'(\hat{N})$.

Stability: If the eigenvalue is positive, then $\hat{N}$ is unstable
" " " " negative, " " " stable

19. $\frac{dC}{dt} = \frac{q}{V}(C_i - C)$

$$T_R = \frac{V}{q}$$

q - inflow/outflow rate

The time required to reduce the initial difference

V - volume of container

$C_0 - C_i$ to e^{-1} .

C_i - initial concentration

$C_0 - C_i$ to e^{-1} .

$$T_R = \frac{12.3 \times 10^6 \text{ m}^3}{220 \text{ L/s}} = 647.1 \text{ days}$$

$$C(T_R) - C_i = e^{-1}(C_0 - C_i)$$

$$C_i = 1.1 \text{ mg/L}$$

$$C(T_R) = e^{-1}(0.3 - 1.1) + 1.1$$

$$C_0 = 0.3 \text{ mg/L}$$

$$C(T_R) \approx 0.806 \text{ mg/L}$$